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Coupled microstructural observations and local texture measurements with an automated crystallographic orientation mapping tool attached to a tem

Gekoppelte mikrostrukturelle Beobachtungen und lokale Gefügemessungen mit einem an ein TEM angeschlossenes, automatisierten System zur kristallographischen Orientierungsbestimmung

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The development of an automated crystal orientation mapping tool attached to a transmission electron microscope is described. The electron beam displacement is controlled by a remote computer while thousands of diffraction patterns are recorded with an external CCD camera and analysed with a dedicated software. The work being mainly concerned with severely deformed metals, spot patterns rather that Kikuchi lines are considered in this approach. The capabilities and the sensitivity of template matching techniques for orientation identification are stressed. To precisely locate the scanned area on TEM pictures, a pseudo-bright field image reconstruction procedure is proposed. A simple procedure to eliminate some orientation ambiguities is described.

Key words: TEM, Spot Patterns, Electron Diffraction, Pattern Recognition

Die Entwickung eines mit einem TEM verbundenen Systems zur automatisierten Bestimmung der kristallographischen Orientirung wird beschieben. Die Verschiebung des Elektronenstrahls wird mittels eines Remote-Computers kontrolliert während tausende Beugungsmuster mittels einer externen CCD-Kamera aufgenommen und mit Hilfe eines geeigneten Programms analysiert werden. Hauptsächlich sind im Rahmen dieser Arbeit stark verformte Metalle - wobei eher Fleckenmuster als Kikuchi-Linien betrachtet werden - betroffen. Die Leistung und die Sensitivität der Erkennungstechnik zur Orientierungsbestimmung werden beschrieben. Zur genauen Bestimmung der gescannten Bereiche von TEM-Bildern wird ein Pseudo-Helligkeitsverfahren zur Bildrekonstruktion vorgeschlagen. Eine einfache Prozedur, um Zweideutigkeiten bei der Orientierung zu eliminieren, ist beschrieben.

Schlüsselworte: TEM, Fleckenmuster, Elektronenbeugung, Mustererkennung

1 Introduction

Transmission electron microscopy remains one of the main characterization tools to infer at a nano scale the physical processes controlling the electrical, magnetic or mechanical properties of most of the materials. Coupled with X-rays or electron detectors and considering the multiple observation modes available (bright/dark fields, diffraction, high resolution, convergent beam, nano beam, etc), it offers a unique source of information on the substructure, misorientations, local chemical gradients, especially for crystalline materials. For the image or chemical aspects, the adequate accessories have reached impressive resolutions and rapidity. By contrast, for crystallographic orientations there are only very few attempts to promote fast and reliable tools to record complete orientation maps at a nano scale. It is worth recalling that it has been demonstrated in the past that the task is fully achievable on TEM [1-3]. The present paper describes the current state of the development of a hardware and software package dedicated to automated crystallographic orientation mapping. The initial objective was to analyse in details the misorientations related to severe plastic deformation. But the device is believed to have high potential fields of utilisation in all metallurgical problems that necessitate nano scale analysis.

2 Orientation identification principle

Basically, the crystallographic orientation is contained in the diffraction pattern. The latter includes Kikuchi lines and/or individual spots. Kikuchi lines are usually preferred presumably because they permit intrinsically a sub-degree orientation resolution. They were made popular with the EBSD (Electron Backscattered Diffraction) attachments that are quite common on modern scanning electron microscopes. Extension for TEMs are also under development (e.g.: Morawiec and co-authors [3,4]). For studies that involve large strains, the spot patterns are preferred. Indeed, they are known to be less sensitive to the internal stresses resulting from severe plastic deformations.

The procedure necessitates (i) fast acquisitions of spot patterns and (ii) reliable orientation identification. This supposes hardware equipments and software developments that will be briefly described before discussing some particular issues of the technique.

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2.1 Spot patterns acquisition

The work was performed on a JEOL 3010 TEM operating at 300 kV. The beam size is decreased to the adequate diameter. A compromise between the beam intensity and spatial resolution must be found. Typically, beam sizes of 20 to 50 nm are used on the microscope that is equipped with a LaB_6 filament. As for EBSD attachments on SEM, the procedure requires a full control of the beam displacement and rapid capture of the diffraction patterns. Both functions are performed by a remote computer equipped with a D/A board that generates a software controlled signal over four channels connected to the electron microscope shift and twist deflecting coils. As in the pionner work of Schwarzer [5], an external CCD camera is used for the acquisition. It collects the patterns at a rate of 25 frames per second through the frame grabber. The 758x486 sensor size is reduced mainly by binning to an area of interest of typically 150x150 pixels [6].

2.2 Identification through template matching

The usual way to extract the orientation from a spot pattern consists in locating the intense spots and to compare their relative positions (angle and distance from the transmitted beam) to values pre-calculated for a given material and known observation conditions. The present approach differs significantly from this scheme [6,7]. Firstly, the spot patterns for sets of Euler's angles covering the Euler space defined by the crystal symmetry are entirely generated and stored in a database. Then, every pattern is compared to the measured one through template matching and the most adequate is selected.

The degree of matching for a given template is characterised by the correlation index that writes:

$$Q(i) = \frac{\sum_{j=1}^{m} P(x_j, y_j) T_i(x_j, y_j)}{\sqrt{\sum_{j=1}^{m} P^2(x_j, y_j)} \sqrt{\sum_{j=1}^{m} T_i^2(x_k, y_j)}}$$

where the pattern is represented by the intensity function P(x,y) while every template *i* is given by the function $T_i(x,y)$. The highest Q value corresponds to the solution. The efficiency of the technique is due to the very limited number *m* of triplets (positions x and y, intensity) that compose the templates: usually around 50 dots are sufficient and therefore only 50 products are computed for every orientation. Moreover, because the values are compared for a given picture, the first part of the denominator is omitted in the calculation. A typical example of pattern matching is shown on *Fig. 1*. The recorded pattern, here for a b.c.c. material, is first numerically filtered to enhance the spot intensity and to locate the transmitted beam. Then, the correlation index is calculated for every possible orientation and the best one is selected. Finally, the solution is superimposed to the pattern on the figure.

The Q values for a given pattern may be visually compared. This is performed on Fig. 2 that shows the so-called correlation indexes map (Q map) related to the pattern of Fig. 1. This is the portion of the stereographic projection that cover all the possible orientations for the given crystal symmetry. For the b.c.c. example considered here, the area is bounded by the [001], [111] and [1-11] directions. The orientation step between two successive templates being one degree, less than 1500 templates are required to cover all the possibilities. On *Fig. 2*, the correlation indexes, divided by the highest one, are converted to grey levels. The selected orientation corresponds to the darkness point on the map.

This map is particularly useful to estimate the reliability of the solution. A single dark area means that the solution is safe; the appearance of extra dark spots on the map indicates that other orientations may be valid and therefore that the solution must be rejected [7]. As a consequence, the reliability may be weighted by the index R that is defined as:

$$R = 100\left(1 - \frac{Q_2}{Q_1}\right)$$

where Q1 and Q2 are the correlation indexes for the two highest and distinct maxima. If R vanishes, the identification

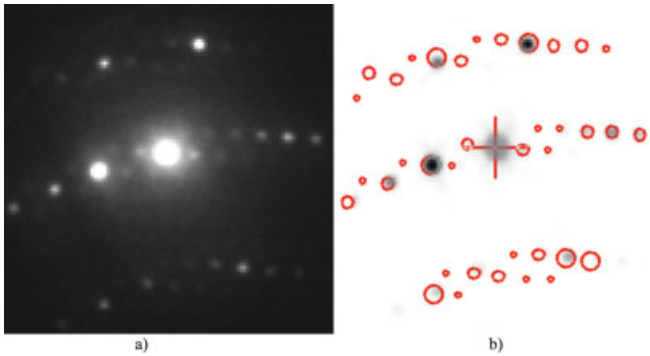

Fig. 1. Example of pattern matching for a random orientation of an α Fe grain a) as acquired, b) filtered image with the selected template superimposed

Abb. 1. Beispiel einer Musterzuordnung für beliebige Orientierung eines α Fe Korns a) wie aufgenommen, b) gefiltertes Bild mit dem ausgewählten Muster überlagert

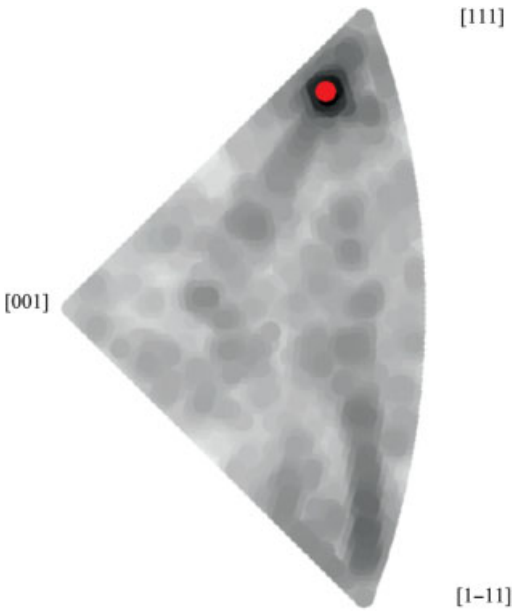

Fig. 2. Correlation indexes map (Q map) with the selected orientation that is close to the [658] axis indicated by the dot

Abb. 2. Abbildung der Korrelationsindizes (Q Abbildung) mit der gewählten Orientierung (mittels eines Punktes angedeuted), die nahe der [658]-Achse ist

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is insecure. This occurs when the pattern is of poor quality either because the intensities of the spots are not high enough or merely because two patterns are superimposed as at grain boundaries. The reliability is also quite poor if the pattern has a 180° ambiguity. This particular problem will be discussed in detail in section 5.

2.3 Software settings optimization

The efficiency of the present procedure to select the adequate orientation is such that part of the identification parameters may be optimized automatically by considering the evolution of Q. This is illustrated with the camera length selection (i.e.: the 'magnification' of the spot pattern). Usually the value is given by the TEM settings but it may vary in a limited way inducing inconsistencies in the solution. Therefore the following procedure proves useful: on Fig. 3, the indexes are calculated for increasing camera lengths. More precisely, the best match is estimated while all the templates are expanded (or contracted) homothetically.

For every camera length some degree of matching may be fortuitously found and the corresponding solution, merely indicated by its number, is given on the plot. These correlations are poor and consequently for two different expansion factors the selected patterns are quite different (*Fig. 3*). The feature of interest is that the real solution is unambiguously recognised thanks to a sharp increase of the index when the camera length approaches the adequate one. For the present case the camera length is equal to 21 (arbitrary units).

This procedure may also be used to select the adequate indexing parameters when a diffraction pattern with unknown observation conditions is analysed. It may be easily understand that the strong sensitivity of the template matching technique on the degree of similarity between the template and the actual diffraction pattern can also be used to distinguish say a b.c.c. pattern from a f.c.c. one. Phase identification performed with the present tool will be discussed in a future paper.

3 Misorientations in severely deformed steel

Figure 4 shows an orientation map and the corresponding highest index values (Q_{max} map) measured for low carbon steel subjected to alternated simple shear up to a cumulated shear strain of 18. A step size of 20 nm was used for the acquisition. Despite the high dislocation density, the orientation is determined in 96 % of the cases.

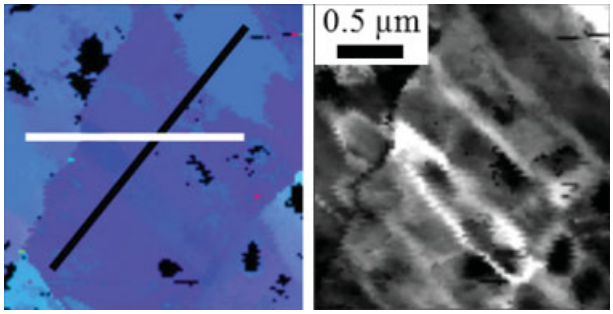

Fig. 4. Orientation and Q_{max} maps for a low carbon steel severely deformed

Abb. 4. Orientierung und Q_{max}-Abbildungen für niedrig gekohlten Stahl unter starker Verformung

As expected, the observed structure is rather complex. Nevertheless, a rather uniform colour is obtained for the orientation map meaning that the normal axis is nearly the same for the whole area. The average orientation is close to the one given on Fig. 2. To analyse more precisely the area, misorientations along the two lines drawn on Fig. 4 are plotted on *Fig. 5*. The substructure appears to be composed of a mixture of cell and grain boundaries that are characterized thanks to the knowledge of the local crystallographic orientation. Of interest is the fact that on the TEM picture the boundaries are quite visible (Fig. 7), but it is hard to notice on the image if a boundary separates cells or grains. As noticed on Fig. 5, the horizontal line crosses a grain boundary (20° misorientation) whereas the other line does not.

It should be mentioned that unfortunately the Q_{max} map does not give a good picture of the sample substructure. Indeed, the correlation index is sensitive to the similarity between the pattern and the selected template but also to the pattern quality that differs substantially from place to place. As a

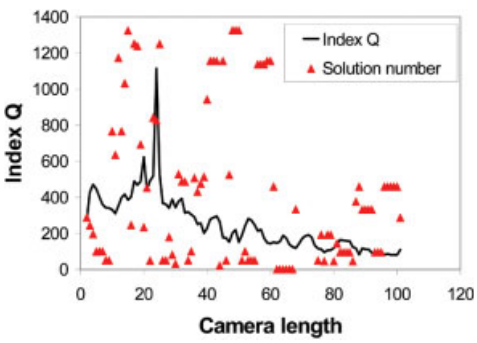

Fig. 3. Correlation indexes in arbitrary units and the corresponding solutions indicated by their numbers for increasing camera lengths. The real camera length coincides with the peak value

Abb. 3. Korrelationsindizes in beliebiger Einheit und die entsprechende Lösung angedeuted durch ihre Nummern für aufsteigende Kammeralängen. Die wahre Kammeralänge stimmt mit dem Peak-Wert überein

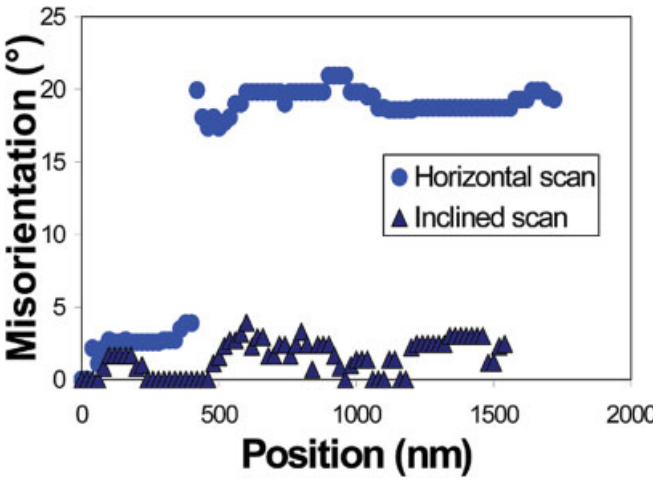

Fig. 5. Misorientations calculated with respect to the first point for the white and black lines drawn on Fig. 4. A grain boundary is detected from the former.

Abb. 5. Berechnete Misorientierung in Bezug auf den ersten Punkt für die in Abb. 4 dargestellten Schwarz-Weiß-Linien. Eine Korngrenze wurde von den vorhergehenden bestimmt.

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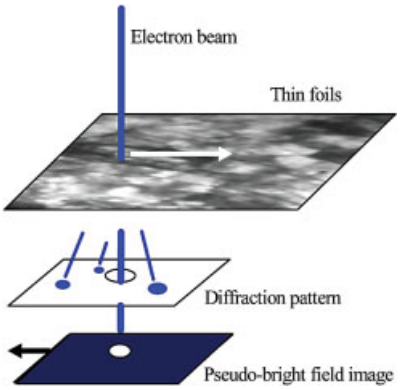

Fig. 6. Pseudo bright field construction. Only the transmitted beam is numerically selected and plotted on the image

Abb. 6. Pseudo-Hellfeld Konstruktion. Nur der übertragene Strahl ist numerisch ausgewählt und auf dem Bild dargestellt

consequence, it is frequently difficult to recognize with the contrast given by the Q indexes, the area that was really scanned. An alternative technique is used for this purpose and is detailed in section 4.

4 Pseudo bright field image

One of the advantages of the present technique with respect to EBSD attachments on SEM is that both orientation map and TEM picture may be recorded together. This supposes that the area scanned for orientation identification is easily located. The task is simplified by the construction of a pseudo bright field image.

As for the standard STEM mode, it is possible to reconstruct a bright field image of the object by plotting the intensity of the transmitted beam while the area of interest is scanned. In the present case, the diffractions patterns being recorded at every step, it is possible to calculate the average intensity in a disc that surrounds the transmitted beam. This is entirely equivalent to the introduction of a virtual aperture in the beam trajectory as shown on *figure 6*.

The pseudo bright field image is obtained by plotting the fluctuating intensity with the same scanning rate and steps than the real one. An example of such a construction is shown in *figure 7* that corresponds to the same area as Fig. 4. It can be noted that the pseudo bright field image is by far better that the Q_{max} one to recognize the scanned area. This type of information is particularly useful to detect any unexpected drift of the beam during the sampling time that is, for 25 frames per second, typically one hour for a full map. Indeed, the beam stability is never perfect mainly because the temperature of the TEM coils is hardly constant. For example, a close inspection of Fig. 7 denotes a small alternating shift of the beam in the lower part of the image.

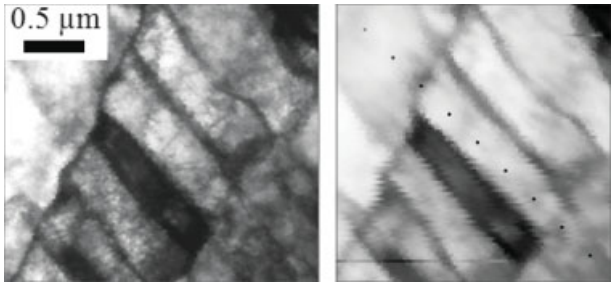

Fig. 7. Real and Pseudo bright field images of the area analysed on Fig. 4

Abb. 7. Reales und Pseudo-Hellfeld Bild des in Abb. 4 analysierten Bereiches

Concerning the spatial resolution, it should be mentioned that for the present reconstructed images, the virtual aperture has to be larger than the transmitted beam. This is because the inner part of the transmitted beam saturates and only the outer part contains contrast information. Therefore, as for STEM mode, it is mainly the beam diameter that delimits the pseudo bright field image resolution.

Of course, the virtual aperture may be displaced on the diffraction pattern and consequently any pseudo dark field image may be constructed in the same way. Of particular interest is the fact that the diffraction patterns being recorded and stored, these reconstructions may be performed off line.

5 Orientation ambiguity

The reliability of the orientation identification from diffraction spot patterns is limited by the well-known 180° ambiguity [8]. A typical example is the [111] pattern for b.c.c. materials that have sixfold symmetry instead of the crystal threefold symmetry. This occurs when there is no spot from the higher order Laue zones. The problem is encountered quite frequently and has to be surmounted by extracting more information from the patterns. Dedicated methods, which concern the equipment itself, are under development but are not implemented yet. Meantime a very simple procedure is used in order to exploit properly the available orientation maps. Basically, the method consists in extracting the adequate information from a collection of patterns. Indeed, if a single a highly symmetric pattern can not be resolved, the association of hundred of adjacent patterns is less problematic.

All the orientations sensitive to the 180° ambiguity are known a priori and have definite signatures. Consequently, it is possible to correct the local orientation by considering the two following facts:

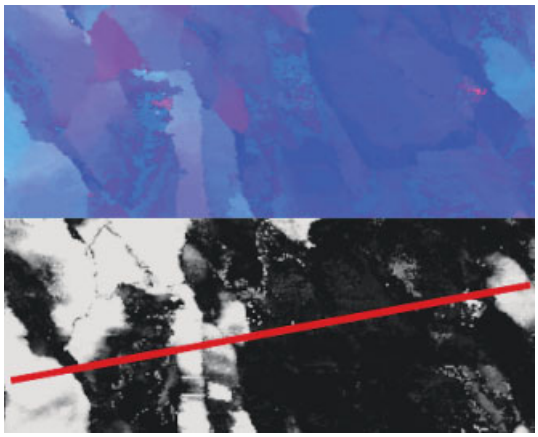

Fig. 8. Orientation map and reliability map (R map) for a low carbon steel. R vanished in the black areas

Abb. 8. Orientierungs- und Vertrauensfeld (R Abbildung) eines niedrig gekohlten Stahls. R verschwindet in den schwarzen Bereichen

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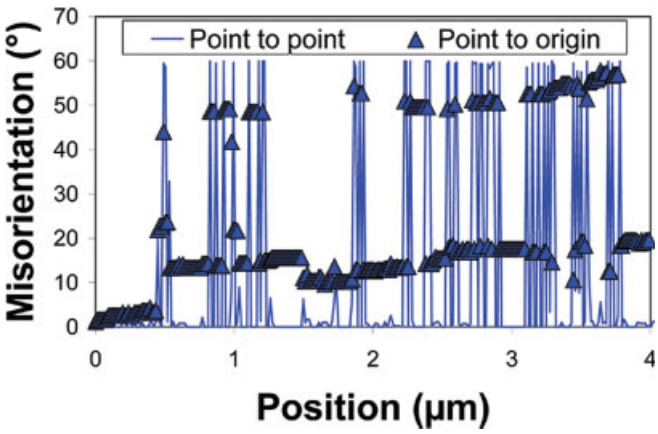

Fig. 9. Misorientations calculated for the line drawn on Fig. 8. A jerky behaviour is observed when the line crosses an unreliable area

Abb. 9. Berechnete Misorientierungen für die in Abb. 8 gezeichnete Linie. Ein ruckartiges Vehalten wird beobachtet, wenn die Linie den unsicheren Bereich durchkreuzt

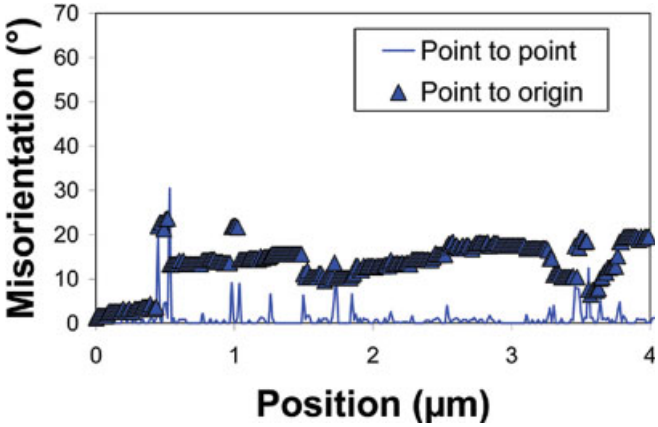

Fig. 10. Same as Fig. 9 but after correction (see the text for details)

Abb. 10. Das Gleiche wie in Abb. 9 aber nach Korrektur (vgl. Text für Details)

- the ambiguities are related to some particular misorientations (e.g. 60° around the normal axis for [111] patterns). The noisy nature of the patterns leads to alternate orientations that are easily detected (Fig. 9)
- small misorientations always exists from one pixel to another and consequently the most ambiguous orientation is surrounded by more reliable ones.

The simplest way to reduce the ambiguity for a given point is to allow the crystal to rotate 180° and to select the orientation that is the closest from the adjacent pixels. In other words the alternated rotations that are related to the ambiguities are merely not permitted. The effective orientation is deduced from the safest point in the neighbourhood (the one that exhibits the highest R value).

Figure 8 gives a typical example of the lack of reliability that arises from the 180° ambiguity in ferrite. The steel was deformed up to a total strain of 7 through Equal Channel Angular Extrusion and large misorientations are recorded (*Fig. 9*). The orientations being close to the [111] axis over the whole area, the reliabilities are poor. This is illustrated by the R map, given also on Fig. 8, where it may be seen that more than half of the identifications are uncertain (black areas). Consequently, the misorientation evolution is jerky with frequent jumps up to 60° (Fig. 9).

For each pixel the selected orientation is rotated 60° around the [111] axis (from symmetry this is equivalent to a 180° rotation) and the one that corresponds to the lowest misorientation with respect to the neighbourhood is selected. Of course, if the resulting misorientation is higher than a threshold value the correction is not performed (grain boundary). The corrected misorientation evolution for the same line as for Fig. 9 is shown on *Fig. 10*. Grain boundaries (point to point misorientation higher than 15°) are now easily distinguished.

6 Conclusions

Template matching algorithms are well adapted for the identification of spot diffraction patterns of randomly oriented crystals. Coupled with a fast image-grabber hardware they constitute an efficient tool for the substructure characterisation of non trivial structures like severely deformed metals. The pseudo bright field image procedure as well as the ambiguity post treatment described in the previous sections constitute valuable complements to the procedure.

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